Glacial Deposits of the Breckenridge Quadrangle

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Introduction

The geologic history of Breckridge, Colorado is a conglomerate of a variety of historic natural events that have happened over millions of years. From early Proterozoic metamorphic rocks, to sandstone deposits from the age of dinosaurs, to glaciers that happened since modern humans existed, the evidence of natural events is embedded in the geologic deposits and formations that we can presently observe. There are two major glaciation events that have shaped the area of present-day Breckenridge, the Pinedale glaciation occurred approximately 30,000 years ago to 10,000 years ago, and the Bull Lake glaciation occurred approximately 150,000 years ago until 130,000 years ago. The variety of rock formations include till deposits, old moraines, outwash deposits, and fan deposits.

Methodology

The models presented for this analysis are manipulated from data taken by the Colorado Geologic Survey, referencing the rock deposits of the Breckenridge Quadrangle covering Summit and Park Counties. Quadrangles are spatially defined by the lines of latitude and longitude. The visuals provided were made using R Studio, to include; a map of all rock types provided by the data, maps of the glaciated rock deposits sectioned by both rock type and glaciation event, and a bar graph demonstrating the percentage of area each rock type covers in the quadrangle. Below is a table detailing each rock type provided by abbreviation in the legends of each map.

af	Artificial fill (Recent)
dt	Dredge tailings (Recent)
ht	Hydraulic tailings (Recent)
mw	Mine waste (Recent)
Qlsh	Historic landslide Cdeposits (Recent)
Qal	Alluvium (Holocene)
Qf	Fan deposits, undivided (Holocene and late Pleistocene)
Qfy	Younger fan deposits (late Holocene)
Qfi	Fan deposits of intermediate age (middle Holo Pleistocene)
Qtn	Neoglacial till (late to middle Holocene)
Qtf	Talus fan deposits (Holocene)
Qti	Inactive talus deposits (Holocene and late Pleistocene)
Qs	Solifluction deposits (Holocene and late Pleistocene)
Qpr	Protalus rampart deposits (Holocene)
Qta	Active talus deposits (Holocene)
Qra	Active rock glacier deposits (late Holocene)
Qr	Inactive rock glacier deposits (early to middle Holocene)
Qls	Landslide deposits, undivided (Holocene to middle Pleistocene)
Qlsy	Younger landslide deposits (late Holocene)
Qlsi	Intermediate age Landslide deposits (middle Holcene to late Pleistocene)
Qlsc	Landslide complex (Holocene and late Pleistocene)
Qlsa	Older landslide deposits (late to middle Pleistocene)
Qc	Colluvium (Holocene and late Pleistocene)
Qrf	Rockfall deposits (Holocene)
Ql	Lacustrine deposits (Holocene and late Pleistocene)
Qac	Alluvium ansd colluvium, undivided (Holocene and late Pleistocene)
Qfp	Fan deposits of Pinedale glaciation (late Pleistocene)
Qop	Pinedale outwash deposits (late Pleistocene)
Qop2	Pinedale outwash deposits of intermediate age (late Pleistocene)
Qop1	Older Pinedale outwash deposits (late Pleistocene)
Qtp	Till of Pinedale glaciation, undivided (late Pleistocene)
Qtp3	Younger till of Pinedale glaciation (late Pleistocene)
Qtp2	Intermediate-age till of Pinedale glaciation (late Pleistocene)
Qtp1	Older till of Pinedale glaciation (late Pleistocene)
Qob	Bull Lake outwash deposits (middle Pleistocene)
Qfb	Bull Lake alluvial fan (middle Pleistocene)
Qtb	Tilt of Bull Lake glaciation (middle Pleistocene)
Qto	Older till, undifferentiated (middle to early Pleistocene)
Qao	Older alluvium, undifferentiated (middle to early Pleistocene)
Qaco	Older alluvium and colluvium (middle and early Pleistocene)
Qgo	Older gravels (middle to early Pleistocene)
QTgg	Gravel of Gold Run (early Pleistocene and late Pliocene)
Tql	Quartz latite porphyry (Eocene?)
Tqmb	Quartz monzonite intrusion breccia (middle Eocene)
Tqm	Quartz monzonite of the Humbug stock (middle Eocene)
Tqp	Quartz monzonite porphyry-megacrystic variety (Eocene)
Tqpm	Quartz monzonite porphyry (Eocene)
Tmp	Hornblende-biotite monzonite porphyry (Eocene)
Kp	Pierre Shale (Late Cretaceous)
Kn	Niobrara Formation (late Cretaceous)
Kb	Benton Shale (Late Cretaceous)
Kd	Dakota Sandstone (Early Cretaceous)
Jme	Morrison Formation and Entrada Sandstone, undivided (Late Jurassic)
Jm	Morrison Formation (Late Jurassic)
Je	Entrada Sandstone (Middle Jurassic)
TRPcm	Chinle Formation (Late Triassic) and Maroon Formation, undivided (Late Pennsylvanian and Early Permian)
PLPmls	Maroon Formation (Late Pennsylvanian and Early Permian)
LPm	Minturn Formation (Middle Pennsylvanian
Dc	Chaffee formation, undivided (Late Devonian)
Dcd	Dyer Dolomite Member
Dcp	Parting Quartzite Member
Om	Manitou Limestone (Early Ordovician)
Csp	Sawatch Quartzite and Peerless Shale, undivided (Late Cambrian)
Cp	Peerless Shale (Late Cambrian)
Cs	Sawatch Quartzite (Late Cambrian)
Yqm	Biotite muscovite quartz monzonite and granite (Middle Protcerozoic)
YXp	Pegmatite, pegmatitic granite, and aplite, undivided (Middle and Early Proterozoic)
YXm	Mafic dikes (Middle and Early Proterozoic)
Yxdi	Diorite (Middle and Early Proterozoic)
Xgg	Granitic gneiss (Early Proterozoic)
Xg	Alaskitic granite (Early Proterozoic)
Xgd	Granodiorite gneiss (Early Proterozoic)
Xmg	Migmatitic biotite gneiss (Early Proterozoic)
Xb	Biotite gneiss (Early Proterozoic)
Xf	Felsic gneiss (Early Proterozoic)

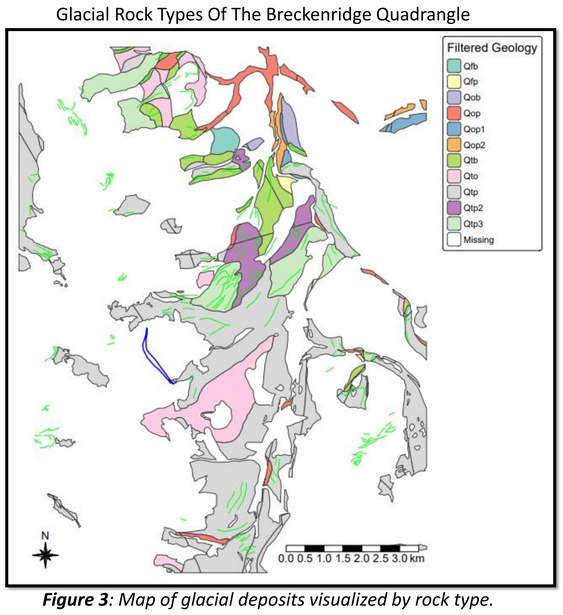

***Figure 3**: Map of glacial deposits visualized by rock type.*

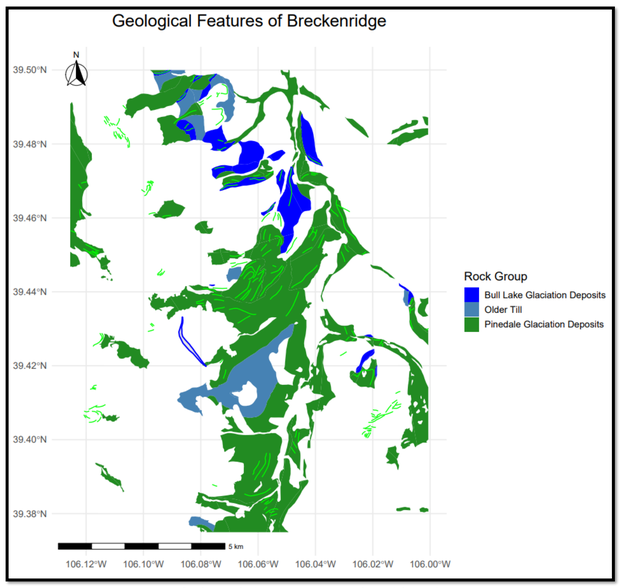

***Figure 2**: Map of glacial deposits visualized by glaciation event.*

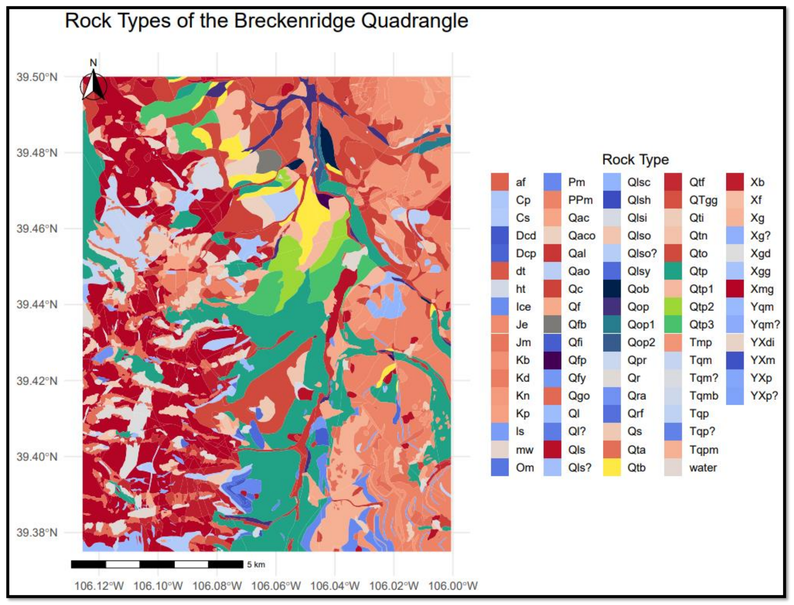

***Figure 1**: Map of all rock deposit types found in the Breckenridge quadrangle.*

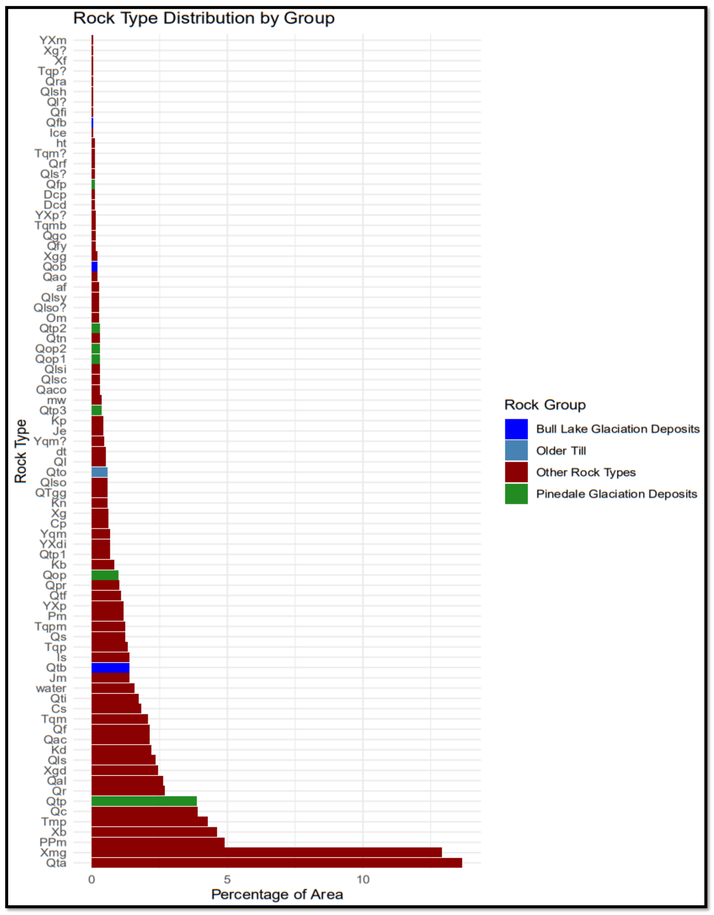

***Figure 4**: Bar graph of percent area covered by each rock type.*

Results and Discussion

The 75 different rock types presented were filtered to display the area being covered by glacial deposits in two of the maps. The sections of rock that were affected by glacial movement (glaciated) on the map are visually significant when compared to the total area of the quadrangle.

The 7th largest rock type by area was Qtp, which was from moraine deposits that occurred during the Pinedale glaciation approximately 20,000 years ago. The Older Till rock deposit (Qto) is poorly sorted larger boulders to smaller rocks that are intermixed from glacial movement. It is older than the Bull Lake glaciation, and its beds can be up to 60 meters thick.

The Talus Deposit (Qta) comprised the most area of the Breckenridge Quadrangle, and it is a fairly recent deposit, from the last 12,000 years (Holocene). However, the second most is the Migmatic Biotite Gneiss (Xmg) which is from the Early Proterozoic timeframe ranging 500 million years to 1 billion years ago.

Contact

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Data Sources

1. https://coloradogeologicalsurvey.org/publications/geologic-map-breckenridge-quadrangle-summit-park-colorado/
2. https://pubs.usgs.gov/pp/0176/report.pdf
3. https://pubs.usgs.gov/circ/1400/circ1400.pdf

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